

A Radar Signal Open-Set Deinterleaving Method Based on Laplacian Pyramid Reconstruction and Adversarial Reciprocal Point Learning

WENBO LI 
YANG-YANG DONG , Member, IEEE
CHUNXI DONG 
RONGHUA GUO 
ZHIYUAN LI 
Xidian University, Xi'an, China

Radar signal deinterleaving is a challenging task in complex electromagnetic environments. This work proposes a method for accurately deinterleaving known and unknown radar radiation sources in an open space using an open-set deinterleaving technique for radar signals based on the combination of Laplacian pyramid reconstruction and reciprocal point adversarial deinterleaving (PR-RPAD). The pulse description graph (PDG) representation of the intercepted pulse description word is first realized by applying a gray matrix symmetric mapping approach based on sliding windows. Second, to improve the visual characterization of the pulse description information of various radar radiation sources, PDG edges and texture structures are amplified and enhanced using the Otsu-threshold-based image amplification technique. Next, a Laplacian-pyramid-based multiresolution feature reconstruction and fusion model is put forth, which uses a jump connection and a multiplication gate to achieve tower reshaping from low-resolution to high-resolution features. The idea of reciprocal points is finally introduced to model the open space to decrease the risk of closed-set deinterleaving through confrontation with known radars. To improve the discriminatory nature of the model against unknown radars and to finish the accurate deinterleaving of radar radiation sources in the open space, an instantiated confrontation enhancement method is used to generate confusing samples for training. In addition to having better performance in closed-set deinterleaving

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Authors' address: Wenbo Li, Yang-Yang Dong, Chunxi Dong, Ronghua Guo, and Zhiyuan Li are with the School of Electronic Engineering, Xidian University, Xi'an 710071, China, E-mail: (lwb20220617@163.com; dongyangyang2104@126.com; chxdong@mail.xidian.edu.cn; 22021110341@stu.xidian.edu.cn; lizzyuan@163.com). (Corresponding author: Yang-Yang Dong.)

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than other deinterleaving techniques (sequential difference histogram (SDIF), pulse repetition interval transform algorithm (PRI-Tran), bidirectional long short-term memory (BLSTM), bidirectional gated recurrent unit (BGRU), and dilated convolutional network (DCN)), the PR-RPAD method also achieves deinterleaving in the open space, which offers great versatility and potential for real-world use in intricate electromagnetic environments.

I. INTRODUCTION

One of the most important components of electronic warfare is electronic support measures (ESM), a device that measures the parameters of radar radiation sources, identifies their properties, and assesses their threat using passive reception and signal processing techniques [1], [2]. The ESM receiver's received radar signal can be viewed as a stream of roughly random radar pulses made up of various radar pulse sequences. The source characteristics and countermeasures can only be determined by splitting the intercepted overlapping pulse streams into pulse sequences that correspond to various radiation sources. Radar signal deinterleaving (RSD) is the process of dividing the overlapping pulse streams into pulse trains that correspond to various transmitters, as seen in Fig. 1.

In the pulsed stream of the intercepted radiation source, the RSD method relies on 5-D characteristics, including time of arrival (TOA), carrier frequency (RF), pulsewidth (PW), pulse amplitude (PA), and direction of arrival (DOA). Schmidt's work [3] on RSD algorithms began as early as 1974. There are fewer published study findings since the associated field is sensitive [4], [5], [6], [7], [8], [9], [10], [11]. Currently, RSD methods can be categorized into those based on traditional pulse description parameters [12] and those based on machine learning [13]. Based on the remarkable achievements of deep learning in the field of image segmentation and open-set recognition (OSR), this article proposes an open-set deinterleaving method for radar signals based on the combination of Laplacian pyramid reconstruction and reciprocal point adversarial deinterleaving (PR-RPAD). The PR-RPAD method realizes the feature characterization from pulse description word (PDW) to pulse description graph (PDG) through the algorithms of symmetric mapping of grayscale matrices and Otsu threshold image enlargement, combined with the Laplacian pyramid multiresolution feature reconstruction and fusion model. The tower reshaping from low-resolution to high-resolution features is accomplished. By introducing the concept of reciprocal points to model the open space and proposing an instantiated adversarial enhancement method based on the adversarial network, under the adversarial mechanism between the reciprocal points and the known radar, the accurate deinterleaving of the known and unknown radar radiation sources in the open space is accomplished. The architecture of the PR-RPAD method is shown in Fig. 2.

In contrast to conventional RSD techniques [5], [14], [15], the PR-RPAD algorithm can adapt to complex pulse repetition interval (PRI) modulation types, performs sequence search without requiring traversal of the data, and

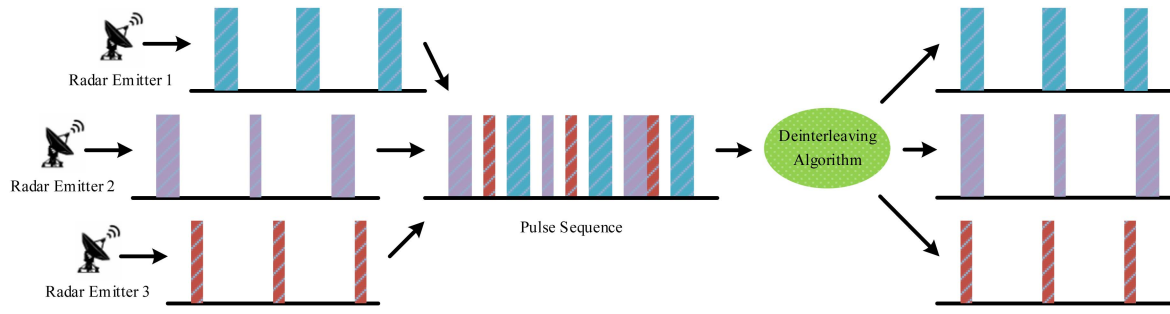


Fig. 1. RSD schematic.

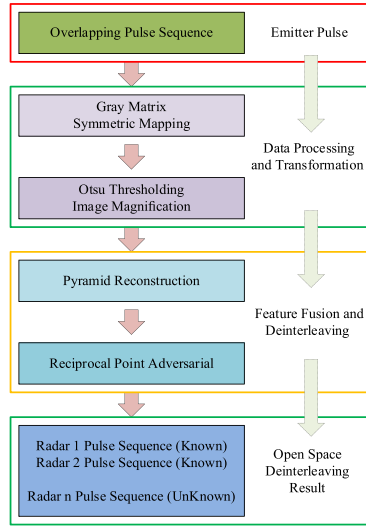


Fig. 2. Architecture of the PR-RPAD algorithm.

does not require multiple rounds of search and merge operations for radar sources with multiple pulses in a single cycle. The PR-RPAD approach eliminates the need to iteratively iterate the input and output data and assign a distinct training network for every radar radiation source, in contrast to neural network and automata-based RSD methods [4], [16], [17], [18]. Using a single network, it is possible to deinterleave overlapping pulse streams from several radiation sources. The PR-RPAD algorithm is capable of accurately deinterleaving known and unknown radar emitters in the open space when compared to semantic segmentation deinterleaving (SSD) (BLSTM, bidirectional gated recurrent unit (BGRU), and DCN) [19] techniques.

The contributions of our work can be summarized as follows.

- 1) *Radar radiation source data processing and conversion:* In this study, the feature characterization from PDW to PDG for the intercepted radar radiation source PDW data is achieved using the sliding-window-based gray matrix symmetric mapping approach. At the same time, the Otsu threshold image amplification technique is used to extend the visual characterization of pulse description information of various radar radiation sources and to achieve the

amplification and enhancement of PDG edges and texture structures.

- 2) *Laplacian pyramid feature reconstruction and fusion:* This study suggests an architecture for multiresolution feature reconstruction and fusion that is based on the Laplacian pyramid. By using jump connections and multiplication gates, the architecture accomplishes the tower reshaping from low-resolution to high-resolution features. It also finishes the varied resolution characterization of PDG edge and texture characteristics of radar radiation sources.
- 3) *Reciprocal point adversarial learning mechanism:* This article models the open space using reciprocal points to lower the risk of closed-set deinterleaving by confronting known radars. At the same time, an instantiated adversarial augmentation method is used to create confusing training samples, improve the model's differentiation against unknown radars, and finish the accurate deinterleaving of radar radiation sources in the open space.

The rest of this article is organized as follows. Section II introduces RSD, image semantic segmentation (ISS), and OSR. In Section III, the problem posed in this article is analyzed and justified. Based on the analysis in Sections II and III, the proposed algorithm in Section IV is constructed—an open-set deinterleaving algorithm for radar signals based on the combination of Laplacian pyramid reconstruction (PR) and reciprocal point adversarial learning. Section V designs relevant experiments and analyzes the experimental results. Finally, Section VI concludes this article.

II. RELATED WORK

A. Radar Signal Deinterleaving

In contemporary electronic warfare, RSD that seeks to distinguish interleaved pulses from various radar radiation sources is crucial [20]. Many attempts have been made in RSD over the past few decades, and these efforts can be divided into two categories: machine-learning-based approaches [13] and approaches based on conventional pulse description parameters [12]. Among other things, pulses from various radar radiation sources are usually separated using intrinsic regularities in the TOA sequence in RSD approaches based on conventional pulse description parameters. One category of machine-learning-based RSD

TABLE I
Characterization of RSD Algorithms

RSD strategy	Typical Algorithm	Characteristic
RSD using standard pulse description parameters	CDIF [24], SDIF [25], PRI-Tran [26], Second-Order DTOA [27], main-lobe pulse correlation [28], Gini impurity [29], semantic codes [30], Markov process [31], and Kalman filter [32]	Features have a simple principle; yet, detection thresholds are rigid and difficult to adapt to complex PRI modulation patterns.
RSD based on machine learning	Clustering methods: K-means [21], DBSCAN [33], AFCM [22], OPTICS [34], AGNES [35], SVMC [36], SOFM [37], and density Dynamic Clustering [23]	Clustering theory is quite interpretative; some methods need a certain number of clusters; it is hard to converge nonconvex datasets, and it is computationally expensive.
	DNNs: RNN, AE, BGRU, BLSTM, DCN, Transformer, and LSTM	Able to adjust to intricate PRI modulation styles; low interpretability.

CDIF: Cumulant Difference Histogram; AGNES: Agglomerative Nesting; SVMC: Support Vector Machine Classifier; SOFM: Self-Organizing Feature Map; LSTM: Long Short-Term Memory;

techniques uses clustering methods, including K-means [21], alternative fuzzy C-means (AFCM) [22], and density dynamic clustering [23], to complete the RSD task. The other category learns discriminative features from tagged pulses using various deep neural network (DNN) types to achieve RSD. In light of this, Table I enumerates common algorithms to RSD and their characteristics.

B. Semantic Segmentation

Deep convolutional neural networks (DCNNs) have demonstrated remarkable efficacy in semantic segmentation and are capable of robustly representing and recognizing objects and materials by leveraging pretrained feature hierarchies. However since spatial deformation invariance is introduced in the feature layer, applying DCNN features to image segmentation results in a decrease in the spatial resolution of the advanced feature representations, and the DCNN features do not adequately represent spatial details. This issue can be resolved by partially recovering the spatial details lost during the maximum pooling process using unpooling and deconvolution, which offers a useful method for visualizing the feedforward model's input characteristics.

Higher spatial resolution information is contained in the feature maps of a DCNN's lower hierarchical structure, although they lack object category specificity. Multiple-level feature maps have been successfully used to classify the response "jet" in semantic segmentation [38], generic border detection, synchronization detection and segmentation, scene recognition, and other applications.

C. Open-Set Recognition

Scheirer et al. [39] originally presented the OSR problem and offered a fundamental structure for training and evaluation following it, drawing inspiration from classifiers with rejection options. Even though OSR is more useful than popular closed-set classifiers, it has been shockingly overlooked in recent years. OSR research can be categorized into two types: generative models and discriminative models.

- 1) *Discriminative models*: RSD techniques based on conventional machine learning were put forth before the application of deep learning. Examples include support vector machines [40], extreme value machines [41], open-set nearest neighbor methods [42], sparse representation OSR methods [43], OpenMax models [44], and K-sigmoid activation methods [45]. The sigmoid function lacks the compact abating property in the aforementioned study [40]. If infinitely distant inputs are introduced to all training data, the aforementioned attribute can be activated; thus, the open space risk is unlimited.
- 2) *Generative models*: In contrast to discriminative models, generative models create unknown or known samples using generative adversarial networks (GANs) [46], autoencoder (AE) [47], and flow-based models [48]. This aids the classifier in determining the decision border between known and unknown samples. The depth distribution of unknown classes throughout the learning process is not taken into account by the aforementioned approaches, which could result in open space hazards.

III. DESCRIPTION AND PROCESSING OF PULSES

The aforementioned analysis shows that conventional deinterleaving algorithms [5], [14], [15] are hard to adapt to complex PRI modulation styles and need a priori knowledge guidance. The deinterleaving methods based on neural networks [4], [16], [17], [18] are too reliant on a priori information and must train the network independently for every kind of radiated source signal. The aforementioned techniques all rely on time data as their RSD input and are unable to adjust to complex regime radar signal deinterleaving situations. Because of this, this work suggests the PR-RPAD algorithm using the multidimensional data of PDW, including DTOA, PW, RF, PA, DOA, etc. Through data imageization mapping and amplification enhancement processing, RSD is transformed into ISS. This section describes and analyzes the PR-RPAD algorithm's pulse characterization, imageization mapping, and amplification augmentation.

A. Description of Pulse Characteristics

The determination of pulse description characteristics is essential for the extraction of each radar's intercepted pulses from the entangled pulse streams of several radar sources and for identifying the radar. Every intercepted pulse's RF, PW, PA, TOA, and DOA are measured by ESM.

TABLE II
Record Sheet for PDW Parameter

Pulse No.	TOA	RF (MHz)	PW (μs)	DOA (°)	PA (dBm)
1	10:01:01.345	3,025	0.75	50.5	-33.5
2	10:01:01.347	9,340	0.32	120.2	-48.5
3	10:01:01.353	2,982	1.51	252.4	-23.3
4	10:01:01.365	3,030	0.71	49.8	-33.9
5	10:01:01.372	2,980	1.52	252.9	-23.2
6	10:01:01.385	3,035	0.72	50.2	-33.7



Fig. 3. Acquisition of PDW data.

As indicated in Table II, every pulse description parameter is documented in a PDW. Pulse-level radar radiometric data are essential for creating an efficient radar library for the ESM and the radar warning system, and PDW data recording helps evaluate the ESM system's effectiveness and grow the electronic intelligence database.

Meanwhile, the radar adjusts the radiation values of each descriptive parameter in response to the different tasks performed. This adjustment, at a certain moment, is manifested in the synergistic cooperation among RF, PW, and PA; during the detection cycle, it is manifested in the potential regularity and temporal sequence of the pulse sequence, including the asymptotics of DTOA and DOA, the group variability of RF and PW, and the antenna scanning characteristics of PA. The descriptive properties of various radar radiation sources vary from one another. Therefore, it is possible to extend the differences between the pulse description parameters of various radar radiation sources in the pulse sequence and improve the connection between different parameter types within a single pulse description parameter by transforming the original pulse description parameters. This also allows for the visualization of the positional relationship between the current pulse and the previous and subsequent pulses.

B. Pulse Conversion Based on Gray Matrix Mapping

To achieve the visual characterization of the radar PDWs (TOA, RF, PW, PA, and DOA), as illustrated in Fig. 3, the DTOA sequence between adjacent pulses is obtained by performing a first-order difference of the intercepted TOA sequence. The DTOA of the first pulse is set to zero, resulting in the new characterization of the TOA sequence

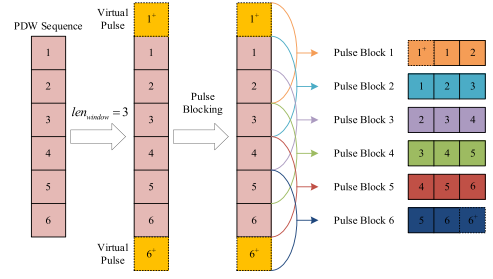


Fig. 4. Pulse data chunking using a fixed window length.

and DTOA sequence. A new description of the radar radiation source's pulse description parameters, namely, DTOA, RF, PW, PA, and DOA, can be derived by combining the dimensional pulse description information of RF, PW, PA, and DOA with the DTOA.

Based on grayscale mapping and the 5-D radar radiation source pulse description parameters (DTOA, RF, PW, PA, and DOA) mentioned earlier, this article suggests a grayscale matrix symmetric mapping method based on a fixed window length to achieve the visual characterization of radar radiation source pulse description parameters, i.e., PDG. This method can extend the differences between various radar radiation source pulse description parameters in the pulse sequence and improve the connection between different parameter types within a single pulse description parameter. In addition, it visualizes the positional relationship between the current pulse, the preceding pulse, and the succeeding pulse. The following are the method's specific steps.

- 1) In the context of radiation source interception, the radar PDW sequence can be written as follows: $PDW = \{PDW_1, PDW_2, \dots, PDW_N\} = [DTOA, RF, PW, PA, DOA]$, where PDW_n is the n th PDW, $PDW_n = [DTOA_n, RF_n, PW_n, PA_n, DOA_n]$, $n \in \{1, 2, \dots, N\}$; N is the sequence length. $PDW_Normalized$ is obtained by normalizing the $DTOA$, RF , PW , PA , and DOA in the PDW , respectively. Each pulse description parameter has a distribution of $[0, 1]$.
- 2) Set window lengths len_{window} and $\{len_{window} | len_{window} = 2k+1, k \in \{0, 1, 2, \dots\}\}$; then, use chunks to process $PDW_Normalized$. When $len_{window} > 1$, fake pulses must be added preceding the start pulse $PDW_Normalized_1$ and following the end pulse $PDW_Normalized_N$ of $PDW_Normalized$. This preserves the consistency of the data length and format following chunking and guarantees that the current pulse is always in the middle of the window length len_{window} . These fake pulses are defined as duplicates of the start and end pulses, and the total number of added pulses is $virtual_{pulse_num} = (len_{window} - 1)/2$. Use Table II data as an example, and then, process $len_{window} = 3$ data chunking, as illustrated in Fig. 4.

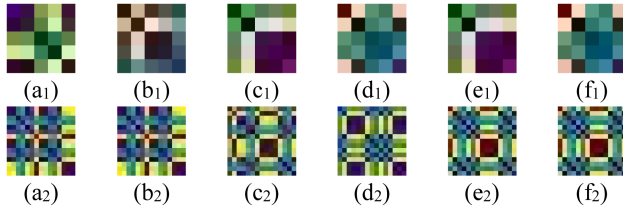


Fig. 5. PDW visual description using symmetric mapping of grayscale matrices with specified window lengths.

3) Data block tiled follows the chunking process. Each of the three pulses that make up Pulse Blocks 1–6 in Fig. 4 comprises parameters, such as DTOA, RF, PW, PA, and DOA. Thus, to create a vector of length 15, a single pulse block is tiled. Using Pulse Block 1 as an example, the following is the expression following spreading:

$$\begin{aligned} \text{PulseBlock}_{1 \text{ Expand}} &= [\text{PDW}_{1+} \quad \text{PDW}_1 \quad \text{PDW}_2] \\ &= [x_1 \quad x_2 \quad \dots \quad x_{15}] \end{aligned} \quad (1)$$

where fake pulse: $\text{PDW}_{1+} = [\text{DTOA}_{1+} \quad \text{RF}_{1+} \quad \text{PW}_{1+} \quad \text{PA}_{1+} \quad \text{DOA}_{1+}]$, real pulse: $\text{PDW}_1 = [\text{DTOA}_1 \quad \text{RF}_1 \quad \text{PW}_1 \quad \text{PA}_1 \quad \text{DOA}_1]$, and $\text{PDW}_2 = [\text{DTOA}_2 \quad \text{RF}_2 \quad \text{PW}_2 \quad \text{PA}_2 \quad \text{DOA}_2]$, $x_i \in [0, 1]$, $1 \leq i \leq 15$.

4) Using a method based on the symmetric mapping of the grayscale matrix, the $\text{PulseBlock}_{1 \text{ Expand}}$ following the aforementioned Pulse Block 1 pulse block tiled is projected to the 2-D numerical matrix $\text{PulseBlock}_{1 \text{ Matrix}}$ using the following expression:

$$\begin{aligned} \text{PulseBlock}_{1 \text{ Expand}} \\ = \begin{bmatrix} y_{1,1} & y_{1,2} & \dots & y_{1,15} \\ y_{2,1} & y_{2,2} & \dots & y_{2,15} \\ \vdots & \vdots & \ddots & \vdots \\ y_{15,1} & y_{15,2} & \dots & y_{15,15} \end{bmatrix} \end{aligned} \quad (2)$$

where $y_{i,j} = \frac{\text{abs}(x_i - x_j)}{x_{\max} - x_{\min}}$, $x_{\max} = \text{Max}[x_1 \quad x_2 \quad \dots \quad x_{15}]$, and $x_{\min} = \text{Min}[x_1 \quad x_2 \quad \dots \quad x_{15}]$.

Using the pulse parameters listed in Table II as an example, the PDG resulting from symmetric mapping of the gray matrix for $\text{len}_{\text{window}} = 1$ and $\text{len}_{\text{window}} = 3$ is displayed in Fig. 5. The PDGs for the $\text{len}_{\text{window}} = 1$ situation are shown in Fig. 5(a₁)–(f₁) with resolution 5×5 . The PDGs are blurred at this resolution, but we can still determine that they come from the same radar radiation source because the feature distributions of the characterization maps in Fig. 5(c₁) and (e₁), as well as in Fig. 5(d₁) and (f₁), are quite comparable. The PDGs with resolution 15×15 for the $\text{len}_{\text{window}} = 3$ condition are shown in Fig. 5(a₂)–(f₂). PDG clarity at this resolution is better than that in Fig. 5(a₁)–(f₁); however, spurious feature scattering and imprecise characterization of the PDG features in Fig. 5(a₂) and (f₂) are caused by the erroneous pulses at both ends of the sequence. In the PDG of the same radar radiation source, there are variations in the feature scatter that are influenced by the surrounding pulses. As well as taking into account the internal

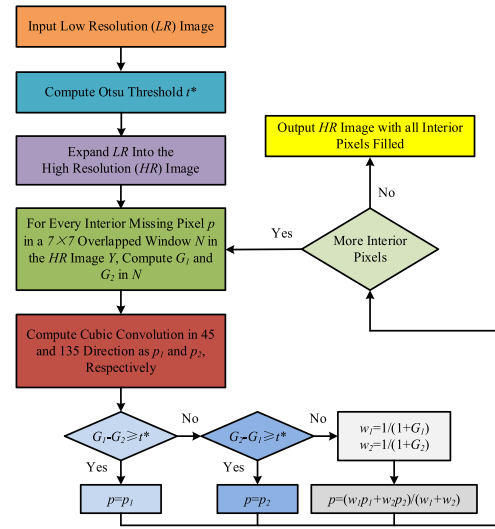


Fig. 6. PDG amplification and enhancement algorithm flowchart based on DCCI_Otsu.

storage capacity constraint following PDG production and the real-time necessity of RSD. Because of this, $\text{len}_{\text{window}}$ cannot be infinitely extended merely to improve clarity and fine feature characterization capabilities and increase the PDG resolution.

C. Magnification and Strengthen of PDG Based on the Otsu Threshold

Drawing from the aforementioned examination of the PDW visual characterization outcomes for symmetric mapping of fixed window length grayscale matrices, the issues of low PDG resolution, blurred features, and absent details in small window lengths, as well as false impulse perturbation, false feature scattering, and imprecise PDG feature characterization in large window lengths, are addressed. In addition, the PDG of the same radar source exhibits variations in feature scattering due to the surrounding impulses. In this article, we propose an Otsu-threshold-based PDG amplification and enhancement algorithm. This algorithm is based on directional cubic convolution interpolation, and the directional cubic convolution interpolation with Otsu thresholding (DCCI_Otsu) method achieves the magnified enhancement of PDG edge and texture structure. In other words, to amplify the PDG regions where there are abrupt changes in the gray values, and to enhance the PDG regions where there are regularities, highlighting the variations, potential combinatorial relationships, and pulse-to-pulse temporal connections among radar radiation source pulse description parameters extends the visual representation of pulse description information of various radar radiation sources and produces more stable and better-magnified images in terms of edge and texture feature retention.

Fig. 6 depicts the operation flow in detail. Together with the pertinent descriptions found in the literature [49], the Otsu threshold t^* is first computed for the low-resolution PDG, LR , followed by the extension of the LR to the high Resolution (HR) PDG, Y , the calculation of the three convolutions in the 45° and 135° directions, respectively,

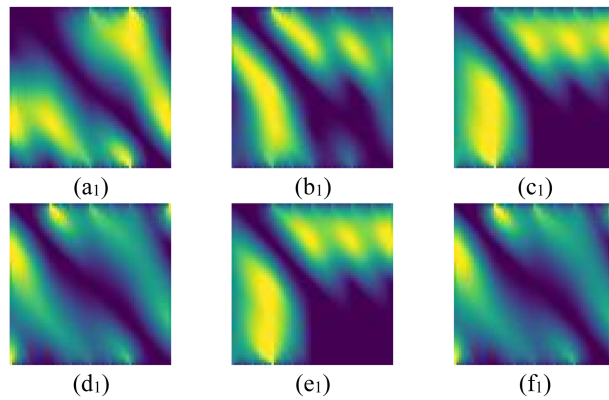


Fig. 7. PDG amplification and enhancement at 65×65 resolution.

and the determination of the missing pixel p , to achieve the PDG amplification and enhancement.

In conclusion, using Fig. 5(a₁)–(f₁) as an example, the original PDGs are amplified and enhanced using the DCCI_Otsu-based PDG amplification and enhancement method; the outcomes are displayed in Fig. 7. The PDG in Fig. 7 is of 65×65 resolution, and when compared to the 5×5 resolution PDG in Fig. 5, it has better characterization findings for the same category of radar radiation source description characteristics, more delicate detail characterization, and much-improved clarity. However, the DCCI_Otsu algorithm's time complexity rises with the PDGs resolution, and the memory needed to store the PDGs increases geometrically. These factors make it unfavorable for the RSD algorithm's eventual implementation.

IV. MODEL CONSTRUCTION

The DCCI-Otsu image amplification and enhancement technique and fixed window length gray matrix symmetric mapping are used to achieve the goal of a high-quality radar radiation source impulse description parameter visualization representation image. In this article, we propose a multiresolution feature reconstruction and fusion model based on the Laplacian pyramid, which mitigates the “spatial semantic uncertainty” phenomenon in the traditional convolutional feature maps, uses jump connections and multiplication gates to realize the tower reshaping from low-resolution to high-resolution features, and completes the multiresolution characterization of PDG edge and texture features. The open space is modeled by introducing the idea of reciprocal points from the standpoint of multi-class integration. Confrontation with known radars lowers the risk of closed-set deinterleaving; to finish the accurate deinterleaving of radar radiation sources in the open space and improve the discriminatory nature of the model against unknown radars, confusion samples are generated for training under the confrontation mechanism between reciprocal points and known radars using an instantiated confrontation enhancement method.

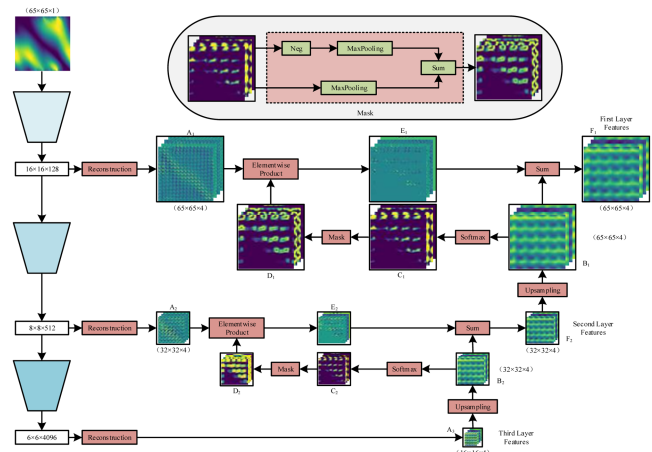


Fig. 8. Architecture of Laplacian pyramid fusion and reconstruction.

A. Laplacian PR Model

A Laplacian-pyramid-based multiresolution feature reconstruction and fusion model is proposed, which draws inspiration from residual networks. Fig. 8 shows the general structure of the Laplacian pyramid multiresolution feature reconstruction and fusion model. The spatial accuracy of the segmentation masks is increased by performing a series of successive reconstructions starting with coarse-scale “low-frequency (it refers to PDG edge information, meaning the region where the PDG gray value quickly changes)” segmentation estimations and adding data from “high-frequency (it refers to PDG’s textural structure information, meaning the region that exhibits regularity)” subbands.

The main body uses the VGG architecture feature extraction network to extract 3-D features, $16 \times 16 \times 128$, $8 \times 8 \times 512$, $6 \times 6 \times 4096$, after first applying the PDG with resolution 65×65 , as illustrated in Fig. 8. The $16 \times 16 \times 128$, $8 \times 8 \times 512$, and $6 \times 6 \times 4096$ feature blocks are then reconstructed into A_1 ($65 \times 65 \times 4$), A_2 ($32 \times 32 \times 4$), and A_3 ($16 \times 16 \times 4$) score maps using the Reconstruction module, which is built using an inverse convolutional network. The number four indicates the number of radar radiation sources. Then, A_3 and F_2 ($32 \times 32 \times 4$) are upsampled, respectively, to achieve the fusion between various resolution pulse description score maps, yielding the upsampled outputs B_2 ($32 \times 32 \times 4$) and B_1 ($65 \times 65 \times 4$). After that, B_2 and B_1 are fed into Softmax to produce C_2 and C_1 , which are subsequently put into the Mask module, which produces D_2 and D_1 by deducting the “low-frequency” components from the high-resolution feature map. D_2 and D_1 are then fed into the Elementwise Product module, where they are multiplied element by element with the original feature maps A_2 and A_1 to produce E_2 and E_1 , or the results of the primary fusion of the features. Finally, B_2 and B_1 are added together with E_2 and E_1 to produce F_2 and F_1 , or the outcomes of the primary fusion of the features. The secondary feature fusion results, or F_2 and F_1 , are then obtained by adding B_2 and B_1 to E_2 and E_1 . The feature reconstruction and fusion at the 1.0, 0.5, and 0.25 PDG scales are finally finished using F_1 , F_2 , and A_3 as the outputs

of the Laplacian pyramid multiresolution reconstruction and fusion module.

B. Open-Set Networks for Reciprocal Point Adversarial Learning

In a set of n labeled samples $\mathcal{D}_L = \{(x_1, y_1), \dots, (x_n, y_n)\}$, there are N known radiation sources ($y_i \in \{1, \dots, N\}$ is the label of x_i) and a greater amount of test data $\mathcal{D}_T = \{t_1, \dots, t_u\}$ ($\{1, \dots, N\} \cup \{N+1, \dots, N+U\}$ is labeled t_i), with U being the number of unknown radiation sources in the real scenario. The open space $\mathcal{O}_k^{\text{pos}}$ from other known radiation sources and the finite unknown space $\mathcal{O}_k^{\text{neg}}$ are the two subspaces of $\mathcal{O}_k$, assuming that the deep embedding space of category k is $\mathcal{S}_k$ and its corresponding open space is $\mathcal{O}_k$, i.e., $\mathcal{O}_k = \mathcal{O}_k^{\text{pos}} \cup \mathcal{O}_k^{\text{neg}}$.

1) *Reciprocal Point Adversarial Learning*: This article defines $\mathcal{D}_u$ as a sample from outside $\mathcal{D}_L$, and $\mathcal{D}_L^k \in \mathcal{S}_k$, $\mathcal{D}_L^{\neq k} \in \mathcal{O}_k^{\text{pos}}$, and $\mathcal{D}_u \in \mathcal{O}_k^{\text{neg}}$ as positive training data, negative training data, and perhaps unknown data, respectively. Furthermore, the reciprocal point $\mathcal{P}^k$ of the category k is thought to be a possible representation of $\mathcal{D}_L^{\neq k} \cup \mathcal{D}_u$. Accordingly, $\mathcal{O}_k$ is nearer the reciprocal point $\mathcal{P}^k$ than $\mathcal{S}_k$, using the following formula:

$$\max \left(\zeta \left(\mathcal{D}_L^{\neq k} \cup \mathcal{D}_u, \mathcal{P}^k \right) \right) \leq d \quad \forall d \in \zeta \left(\mathcal{D}_L^k, \mathcal{P}^k \right) \quad (3)$$

where $\zeta(\cdot, \cdot)$ represents all the sample distance between the computational sets.

The reciprocal points and the associated known categories have a dyadic relationship that can be used to execute RSD based on the aforementioned equation. To be more precise, a deep embedding function $\mathcal{C}$ with a learnable parameter θ can optimize the reciprocal points of a category and represent them with m dimensions. Combining the Euclidean distance d_e with the dot product d_d yields the distance $d(\mathcal{C}(x), \mathcal{P}^k)$, given a sample x and a reciprocal point $\mathcal{P}^k$

$$d_e(\mathcal{C}(x), \mathcal{P}^k) = \frac{1}{m} \cdot \|\mathcal{C}(x) - \mathcal{P}^k\|_2^2 \quad (4)$$

$$d_d(\mathcal{C}(x), \mathcal{P}^k) = \mathcal{C}(x) \cdot \mathcal{P}^k \quad (5)$$

$$d(\mathcal{C}(x), \mathcal{P}^k) = d_e(\mathcal{C}(x), \mathcal{P}^k) - d_d(\mathcal{C}(x), \mathcal{P}^k). \quad (6)$$

The aforementioned formula is used to estimate the difference between the embedded feature $\mathcal{C}(x)$ and the reciprocal points of the known category to identify the category to which it belongs. The process is then normalized using the Softmax function based on the property that the total is 1, yielding the final classification probability

$$p(y = k | x, \mathcal{C}, \mathcal{P}) = \frac{e^{\gamma d(\mathcal{C}(x), \mathcal{P}^k)}}{\sum_{i=1}^N e^{\gamma d(\mathcal{C}(x), \mathcal{P}^i)}} \quad (7)$$

where the ease of distance-probability conversion is controlled by the hyperparameter γ .

To optimize the learnable parameter θ in the deep embedding function $\mathcal{C}$, the reciprocal point classification loss

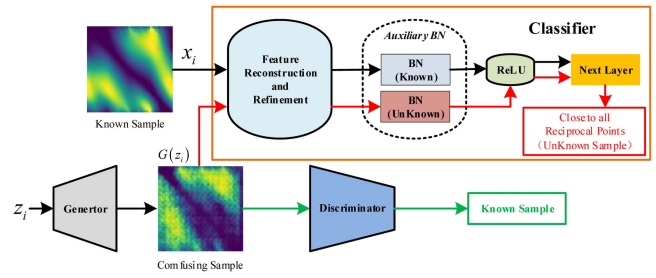


Fig. 9. Network framework for the reciprocal point adversarial learning open set.

based on the true category k negative logarithmic probability is minimized. In addition, the empirical classification risk is decreased by minimizing the following reciprocal point classification loss:

$$\mathcal{L}_c(x; \theta, \mathcal{P}) = -\log p(y = k | x, \mathcal{C}, \mathcal{P}). \quad (8)$$

Apart from categorizing the known categories, the following equation illustrates how the known and unknown spaces are divided by optimizing the distance between the reciprocal points and the matching training samples:

$$\arg \max_{f \in \mathcal{H}} \{ \zeta(\mathcal{D}_L^k, \mathcal{P}^k) \}. \quad (9)$$

The formulation that follows demonstrates how the complementary nature of $\mathcal{S}_k$ and $\mathcal{O}_k$ is used to indirectly constrain the open space by limiting the distance between $\mathcal{S}_k$ and the reciprocal point $\mathcal{P}^k$ to be less than R :

$$\mathcal{L}_o(x; \theta, \mathcal{P}^k, R^k) = \max(d_e(\mathcal{C}(x), \mathcal{P}^k) - R, 0) \quad (10)$$

where R is a learnable parameter; a wider range of non- k samples is obtained using the Euclidean distance.

The total loss function in reciprocal point adversarial learning integrates (8) and (10) and can manage both the open space risk and the empirical classification risk

$$\mathcal{L}(x, y; \theta, \mathcal{P}, R) = \mathcal{L}_c(x; \theta, \mathcal{P}) + \lambda \mathcal{L}_o(x; \theta, \mathcal{P}, R) \quad (11)$$

where θ , $\mathcal{P}$, and R are the learnable parameters and λ is the open space risk module weight.

2) *Instantiate Adversarial Enhancement*: Although such classifiers can differentiate unknown distributions without any prior knowledge of the unknown data, they are vulnerable to perplexing samples produced by basic generators. To enhance the classifier's discriminative capacity for unknown distributions, a generative network is employed to provide more data as confusing samples (CSs) for unknown data $\mathcal{D}_U$.

In this research, rather than recovering known samples $\mathcal{S}_k$, we employ a generator to recover partial CS from the global open space $\mathcal{O}_G$. An adversarial enhancement framework comprising a discriminator D , a generator G , and a classifier C with a deep embedding function $\mathcal{C}$ is created and shown in Fig. 9. The classifier C is used to discriminate the likelihood that a sample belongs to a known class. The latent variable z in the prior distribution $P_{\text{pri}}(z)$ is mapped to the output $G(z)$ by the generator G . After that, the discriminator D is tuned to distinguish between

true and generated samples using $\{z_1, \dots, z_n\}$ in $P_{\text{pri}}(z)$ and known samples $\{x_1, \dots, x_n\}$.

In this study, we propose an adversarial process between reciprocal points and known classes that motivates the generator to generate samples close to each open space $\mathcal{O}_k$ center $\mathcal{P}^k$, i.e., the generated PDG is motivated to be close to the global open space $\mathcal{O}_G$. Instead, to confuse the discriminator D , the generator G anticipates samples that are more similar to the recognized categories. When the two adversarial methods mentioned previously are combined, the generator G is optimized using the following formula:

$$\max_G \frac{1}{n} \sum_{i=1}^n [\log D(G(z_i)) + \beta \cdot H(z_i, \mathcal{P})] \quad (12)$$

where $H(z_i, \mathcal{P}) = -\frac{1}{N} \sum_{k=1}^N S(z_i, \mathcal{P}^k) \cdot \log(S(z_i, \mathcal{P}^k))$ is the information entropy function and β is the hyperparameter that regulates the information entropy's loss weight.

To train the feature space, utilize the generated samples as unknown data $\mathcal{D}_U$. Then, optimize the classifier C using the following formula to minimize the open space $\mathcal{O}_G$ size, which lowers the open space risk:

$$\min_C \frac{1}{n} \sum_{i=1}^n [\mathcal{L}(x_i, y_i) - \beta \cdot H(z_i, \mathcal{P})] \quad (13)$$

where $\mathcal{L}$ represents adversarial learning overall loss at reciprocal points.

The aforementioned process, the auxiliary batch normalization method, stops the CS from having a detrimental effect on the discriminations between known categories. Finally, mutually reinforcing joint training between the discriminator D and the classifier C and the CS generator G using an alternating algorithm optimizes the aforementioned goals. By training the classifier C with the known categories after the CS trains the classifier C , the classifier C is encouraged to concentrate on the known categories and compensates for the CS bias.

C. Pyramid Reconstruction and Reciprocal Point Adversarial Deinterleaving

The general flow of the reciprocal point adversarial learning and Laplacian PR-based radar signal open-set deinterleaving approach presented in this article is illustrated in Fig. 10. Using a three-level Laplacian PR and fusion model, the PR-RPAD model uses the PDG with resolution size 65×65 as input and produces feature score maps with resolutions $65 \times 65 \times N$, $32 \times 32 \times N$, and $16 \times 16 \times N$ from top to bottom. The number of known radar sources is N . Consequently, reciprocal point adversarial learning generates three learning processes, each of which is optimal under the loss function displayed in (11). The overall loss function of the PR-RPAD method is obtained by taking a weighted sum of the learning losses of the various processes

$$\mathcal{L}_{\text{PR-RPAD}} = \alpha_1 \mathcal{L}_{65 \times 65 \times N} + \alpha_2 \mathcal{L}_{32 \times 32 \times N} + \alpha_3 \mathcal{L}_{16 \times 16 \times N} \quad (14)$$

where $\mathcal{L}_{65 \times 65 \times N}$, $\mathcal{L}_{32 \times 32 \times N}$, and $\mathcal{L}_{16 \times 16 \times N}$ are the reciprocal point adversarial learning loss functions displayed in various processes for (11); α_1 , α_2 , and α_3 are the loss function weight hyperparameters.

V. EXPERIMENTAL DETAILS

Simulation tests are carried out in this section to verify the PR-RPAD algorithm deinterleaving capability in the presence of known and unknown mixed radar radiation source signals. The configuration of the measurement interception scenario for the simulation experiments is described in Section V-A. Details of the simulation data parameter settings and experimental distribution are provided in Section V-B. Evaluation metrics, control experiments, training methodology, and dataset setup are all covered in Section V-C. In Section V-D, the deinterleaving outcomes of the PR-RPAD algorithm are described and examined.

A. Data Simulation

This work establishes two radar radiation source measurement interception scenarios based on the PR-RPAD algorithm model characteristics and the description of the radar radiation source pulse parameters in Section III-A. Scenarios 1 and 2 include the interception of radar radiation source measurements in closed and open spaces.

In Scenario 1, the intercept pulse data simulation using (a), (b), and (c) is measured using three radar radiation sources. The simulation's difficulty and data complexity are growing steadily. The details are listed as follows.

- Measurements intercept pulse data simulation for single-function radar radiation sources. All of the radar radiation sources in these simulation data are single-function radars, each of which has a single set of radiation parameters. The data only show three PRI modulation styles: Fixed, Staggered, and D&S.
- Measurements intercept pulse data simulation for multifunction radar radiation sources. Both single-function and multifunction radars are depicted in these simulated data; the multifunction radars are freely deployable based on the mission mode and have two or more sets of radiation parameters. The data only show three PRI modulation styles: Fixed, Staggered, and D&S. Simulated data simultaneously show radiated sources of the same type of radar with the same RF, PW, and PRI, with the only differences being in DOA and PA.
- Measurements intercept pulse data simulation for radar radiation sources under jittered PRI. Both single-function and multifunction radars are depicted in these simulated data; the multifunction radars are freely deployable based on the mission mode and have two or more sets of radiation parameters. The data exhibit four PRI modulation styles: Fixed, Staggered, Jittered, and D&S. Simulated data simultaneously show radiated sources of the same type of radar with the same RF, PW, and PRI, with the only differences being in DOA and PA.

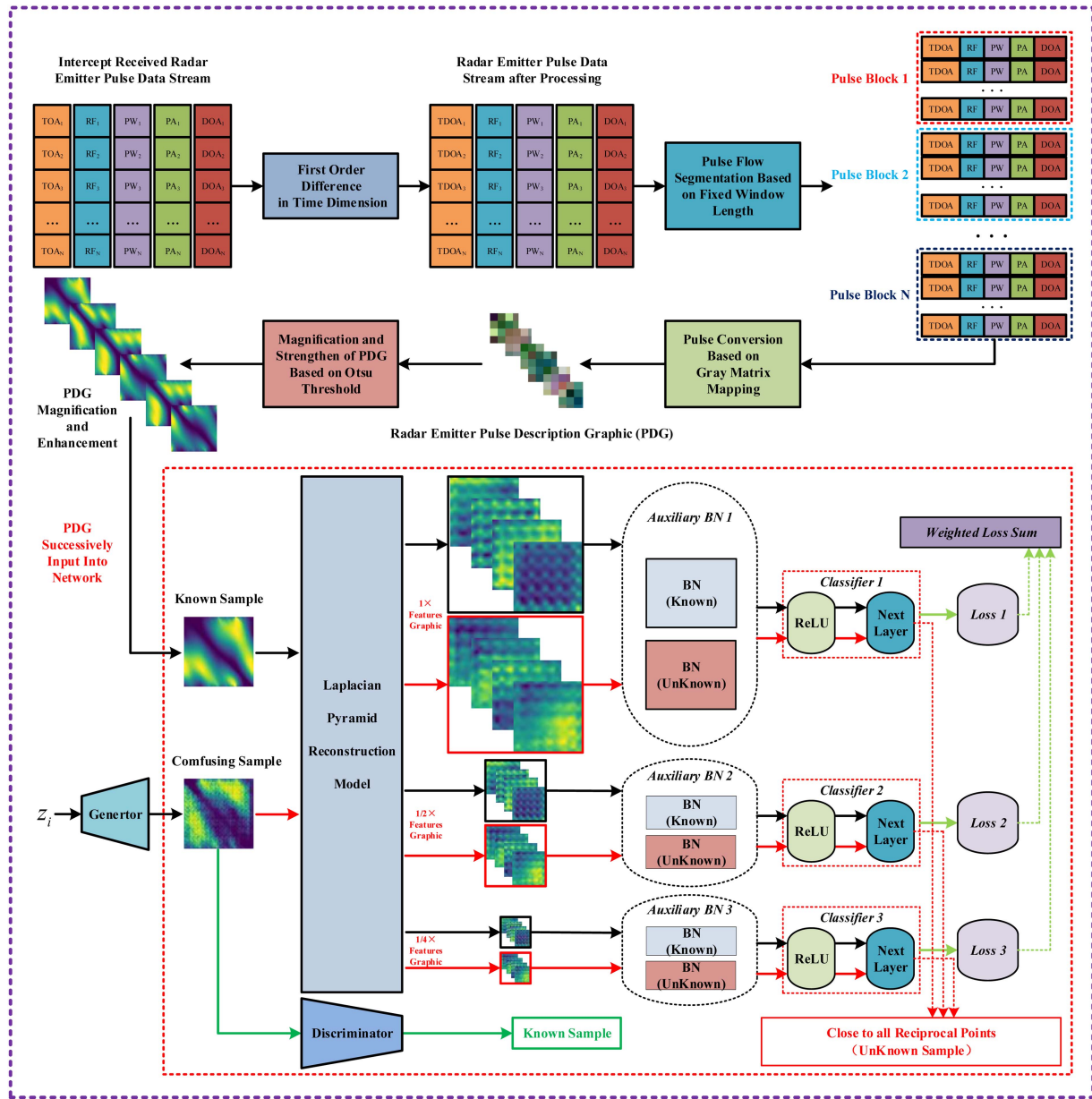


Fig. 10. PR-RPAD method.

In scenario 2, the complexity and difficulty of the data gradually rise when three different types of radar radiation sources—(d), (e), and (f)—are put up to mimic the intercepted pulse data. Corresponding to (a), (b), and (c) in Scenario 1, respectively, and to test the PR-RPAD algorithm's open-set deinterleaving capability, many radar radiation sources were randomly assigned as unknown radiation sources.

It is necessary to take into account pulse loss and random noise brought on by modifications to the measurement interception environment in the data simulation above. The pulse loss rate is denoted as follows:

$$\rho_l = \frac{N_{\text{loss}}}{N_{\text{sum}}} \quad (15)$$

where N_{sum} represents the total number of pulses from the radiation source and N_{loss} represents the number of pulses lost from the source.

ρ_n is the ratio of the average number of target pulses to the number of random noise, i.e.,

$$\rho_n = \frac{N_{\text{noise}}}{N_{\text{average}}} \quad (16)$$

where N_{average} is the average number of pulses from the radiation source and N_{noise} is the number of random noises. $\rho_n/(\rho_n + C)$ can be used to determine the ratio of random noise to total pulses, while C represents the number of radiation sources.

TABLE III
Radar Radiation Source Parameters for Experiment 1

Tag	RF (MHz)	PW (μ s)	PRI (μ s)	PA (dBm)	DOA ($^{\circ}$)
1	9650–9950 F	7.2–8.6 F	61–92 S	–45 to –8 M	79– 84
2	9820–10 200 A	1.2–3.8 S	28–32 DS	–42 to –35 P ₁	76– 81
3	1205–1450 S	4.2–5.7 A	59.1 F	–54 to –49 P ₂	80– 81
4	8200–12 300 F	1.4–6.2 F	18.7–48.7 S	–51 to –43 P ₁	82– 88
0	Clutter/noise/jamming pulse				

TABLE IV
Radar Radiation Source Parameters for Experiment 2

Tag	RF (MHz)	PW (μ s)	PRI (μ s)	PA (dBm)	DOA ($^{\circ}$)
1	9650–9950 F	7.2–8.6 F	61–92 S	–45 to –38 M	79– 84
2	9820–10 200 A	1.2–3.8 S	28–32 DS	–42 to –35 P ₁	76– 81
3	9650–9950 F	7.2–8.6 F	61–92 S	–40 to –33 M	75– 80
4	1205–1450 S	4.2–5.7 A	59.1 F	–54 to –49 P ₂	80– 81
5	8200–10 100 F	1.4–4.5 F	18.7–28.7 S	–51 to –43 P ₁	82– 88
	10 200–12 300 S	3.2–6.2 A	29–68.7 DS	–44 to –37 P ₁	82– 88
0	Clutter/noise/jamming pulse				

B. Design of Experiments

The experiments that follow are intended to confirm the PR-RPAD algorithm viability and the deinterleaving's performance using the six simulated data situations in the two scenarios mentioned earlier. Among them, Experiments 1–3 match the data simulation's (a)–(c); Experiments 4–6 match the data simulation's (d)–(f). All experimental data ρ_l and ρ_n are chosen at random from a predetermined range. The amplitude data simulation is produced in the data-generating process of the aforementioned experiment by adjusting the radar radiation source antenna's scanning pattern and the range of its amplitude value ups and downs. In addition, mechanical scanning is M, 1-D phase scanning is P₁, and 2-D phase scanning is P₂. The full design of the experimental data is as follows (Known: K, Unknown: U; Fixed: F, Agile: A, Stagger: S, D&S: DS, and Jitter: J).

Experiment 1: A single-function radar radiation source in closed-set space measures the deinterleaving of intercepted pulse signals. The parameters for the radar radiation source are configured as indicated in Table III, where $0 \leq \rho_l \leq 0.5$ and $0 \leq \rho_n \leq 0.5$.

Experiment 2: A multifunction radar radiation source in closed-set space measures the deinterleaving of intercepted pulse signals. The parameters for the radar radiation source are configured, as indicated in Table IV, where radiation sources 1 and 3 are radiation sources of the same type and

TABLE V
Radar Radiation Source Parameters for Experiment 3

Tag	RF (MHz)	PW (μ s)	PRI (μ s)	PA (dBm)	DOA ($^{\circ}$)
1	9650–9950 F	7.2–8.6 F	61–92 S	–45 to –38 M	79– 84
2	9820–10 200 A	1.2–3.8 S	28–32 DS	–42 to –35 P ₁	76– 81
	10 800–11 200 F	2.2–6.8 A	18–46 J	–45 to –38 P ₁	76– 81
3	9650–9950 F	7.2–8.6 F	61–92 S	–40 to –33 M	75– 80
4	1205–1450 S	4.2–5.7 A	59.1 F	–54 to –49 P ₂	80– 81
5	8200–9100 F	1.4–4.5 F	18.7–28.7 S	–51 to –43 P ₁	82– 88
	9120–10 200 S	3.2–5.2 A	29–68.7 DS	–44 to –37 P ₁	82– 88
	11 200–12 300 A	4.1–6.4 S	21.4–34.7 J	–48 to –45 P ₁	82– 88
6	5400–5870 S	1.1–11.2 F	990–1007 F	–41 to –34 P ₂	65– 69
7	3100–3490 A	22.4– 29.1 F	857–1250 J	–39 to –32 P ₁	85– 86
0	Clutter/noise/jamming pulse				

TABLE VI
Radar Radiation Source Parameters for Experiment 4

Class	Tag	RF (MHz)	PW (μ s)	PRI (μ s)	PA (dBm)	DOA ($^{\circ}$)
K	1	9650– 9950 F	7.2– 8.6 F	61–92 S	–45 to –38 M	79– 84
U	2	9820– 10 300 A	1.2– 3.8 S	28–32 DS	–42 to –35 P ₁	79– 81
K	3	1205– 1450 S	4.2– 5.7 A	59.1 F	–54 to –49 P ₂	80– 81
K	4	8200– 12 300 F	1.4– 6.2 F	18.7– 48.7 S	–51 to –43 P ₁	82– 88
K	0	Clutter/noise/jamming pulse				

radiation source 5 is a multifunction radiation source. $0 \leq \rho_l \leq 0.5$ and $0 \leq \rho_n \leq 0.5$.

Experiment 3: A radar radiation source with jittered PRI in closed-set space measures the deinterleaving of intercepted pulse signals. The parameters for the radar radiation source are configured, as indicated in Table V, where radiation sources 1 and 3 are radiation sources of the same type and radiation sources 2 and 5 are multifunction radiation sources. $0 \leq \rho_l \leq 0.5$ and $0 \leq \rho_n \leq 0.5$.

Experiment 4: A single-function radar radiation source in an open-set space measures the deinterleaving of intercepted pulse signals. The parameters for the radar radiation source are configured, as indicated in Table VI, where radiation source 2 is specified as an unknown radiation source; $0 \leq \rho_l \leq 0.5$, and $0 \leq \rho_n \leq 0.5$.

Experiment 5: A multifunction radar radiation source in an open-set space measures the deinterleaving of intercepted pulse signals. The parameters for the radar radiation

TABLE VII
Radar Radiation Source Parameters for Experiment 5

Class	Tag	RF (MHz)	PW (μ s)	PRI (μ s)	PA (dBm)	DOA ($^{\circ}$)
K	1	9650– 9950 F	7.2– 8.6 F	61– 92 S	–45 to –38 M	79– 84
U	2	9820– 10 200 A	1.2– 3.8 S	28– 32 DS	–42 to –35 P ₁	76– 81
U	3	9650– 9950 F	7.2– 8.6 F	61– 92 S	–40 to –33 M	75– 80
K	4	1205– 1450 S	4.2– 5.7 A	59.1 F	–54 to –49 P ₂	80– 81
K	5	8200– 10 100 F	1.4– 4.5 F	18.7– 28.7 S	–51 to –43 P ₁	82– 88
		10 200– 12 300 S	3.2– 6.2 A	29– 68.7 DS	–44 to –37 P ₁	82– 88
K	0	Clutter/noise/jamming pulse				

TABLE VIII
Radar Radiation Source Parameters for Experiment 6

Class	Tag	RF (MHz)	PW (μ s)	PRI (μ s)	PA (dBm)	DOA ($^{\circ}$)
K	1	9650– 9950 F	7.2– 8.6 F	61– 92 S	–45 to –38 M	79– 84
U	2	9820– 10 200 A	1.2– 3.8 S	28– 32 DS	–42 to –35 P ₁	76– 81
		10 800– 11 200 F	2.2– 6.8 A	18– 46 J	–45 to –38 P ₁	76– 81
U	3	9650– 9950 F	7.2– 8.6 F	61– 92 S	–40 to –33 M	75– 80
K	4	1205– 1450 S	4.2– 5.7 A	59.1 F	–54 to –49 P ₂	80– 81
K	5	8200– 9100 F	1.4– 4.5 F	18.7– 28.7 S	–51 to –43 P ₁	82– 88
		9120– 10 200 S	3.2– 5.2 A	29– 68.7 DS	–44 to –37 P ₁	82– 88
		11 200– 12 300 A	4.1– 6.4 S	21.4– 34.7 J	–48 to –45 P ₁	82– 88
U	6	5400– 5870 S	1.1– 11.2 F	990– 1007 F	–41 to –34 P ₂	65– 69
K	7	3100– 3490 A	22.4– 29.1 F	857– 1250 J	–39 to –32 P ₁	85– 86
K	0	Clutter/noise/jamming pulse				

source are configured, as indicated in Table VII, where radiation sources 1 and 3 are radiation sources of the same type, radiation source 5 is a multifunction radiation source, and radiation sources 2 and 3 are specified as unknown radiation sources; $0 \leq \rho_l \leq 0.5$ and $0 \leq \rho_n \leq 0.5$.

Experiment 6: A radar radiation source with jittered PRI in an open-set space measures the deinterleaving of intercepted pulse signals. The parameters for the radar radiation source are configured, as indicated in Table VIII, where radiation sources 1 and 3 are radiation sources of the same type, radiation sources 2 and 5 are multifunction radiation sources, and radiation sources 2, 3, and 6 are specified as unknown radiation sources; $0 \leq \rho_l \leq 0.5$ and $0 \leq \rho_n \leq 0.5$.

C. Datasets, Training Methods, Control Experiments, and Performance Metrics

1) *Dataset:* In this article, two types of datasets—a mixed dataset and a particular intercepted environment dataset—are produced during the experiments. In each experiment, there is only one hybrid dataset, which is made up of PDW sequences with a sample size of 200 and a length of 400 per sample for training and validating RSD models. ρ_l and ρ_n are split among $[0, 0.5]$. There are two categories into which specific-intercept scenario datasets can be divided: ① there are five datasets with measurements and intercepts of reconnaissance receivers in good condition (with a pulse loss rate of $\rho_l = \{0\}$) and interception environments that are gradually getting worse (with noise ratios of $\rho_n = \{0.1, 0.2, 0.3, 0.4, 0.5\}$) and ② in interception status and reconnaissance receiver measurements, the interception environment progressively deteriorates, and the impulse loss rate equals the noise ratio, i.e., five datasets $\rho_l = \rho_n = \{0.1, 0.2, 0.3, 0.4, 0.5\}$. The aforementioned ten datasets, each including 100 PDW sequences with a length of 400, are used to test the RSD algorithm's performance under various measurement interception conditions.

2) *PR-RPAD Model Training Approach:* ① *Closed-set scenario training:* The training dataset, validation dataset, and test dataset are separated in an 8:1:1 ratio. The PR-RPAD model was trained in an environment with Windows 10 22H2, Python 3.8.16 64-bit, and Torch 2.0.1+cu117. With 100 iterations and a learning rate of 0.01, the stochastic gradient descent (SGD) optimizer was employed. No instantiation enhancement mechanism is added, and GANs are not required to synthesize obfuscated samples.

② *Open-set scenario training:* Within the hybrid dataset, the known radiation source training, validation, and test datasets (Test Dataset) are split in an 8:1:1 ratio. The unknown radiation source data are set as the unknown class dataset (Out Dataset). The Test Dataset and the Out Dataset are utilized for evaluating the model after training. The PR-RPAD model was trained in an environment with Windows 10 22H2, Python 3.8.16 64 bit, and Torch 2.0.1+cu117. With 100 iterations and a learning rate of 0.01, the SGD optimizer was employed. The confused sample generation network learning rate is 0.0002, and the Adam optimizer was employed.

In conclusion, the real-time RSD requirement and the internal storage capacity limitation following PDG generation are taken into account during the PR-RPAD model training and testing process. The PDGs were generated in all tests under the $len_{window} = 1$ condition that the model inputs were PDG-enhanced and magnified images with a 65×65 resolution. To achieve RSD with a reduced time cost and less memory space, the captured interleaved pulse data are fed as a stream and processed sequentially, finally freeing the cache and outputting the result.

3) *Control Experiments:* ① *Closed-set scenario:* BLSTM, BGRU, DCN, SDIF, and PRI-Tran.

② *Open-set scenario:* SDIF and PRI-Tran.

The forward and reverse DTOA information is equal in RSD. To achieve RSD, the literature [19] uses bidirectional

TABLE IX
Experiment 1: Analysis of Model Performance on a Mixed Dataset
(Closed Set)

Metrics Method	$ACC_{closed-set}$	MR	MP	MIOU
SDIF	44.889	38.872	48.514	24.974
PRI-Tran	38.459	31.818	39.523	18.440
BLSTM	58.566	57.361	57.628	40.626
BGRU	60.908	60.314	60.838	43.742
DCN	44.737	39.180	42.073	25.299
PR-RPAD	100.000	100.000	100.000	100.000

Bold values indicate the best performance.

TABLE X
Experiment 2: Analysis of Model Performance on a Mixed Dataset
(Closed Set)

Metrics Method	$ACC_{closed-set}$	MR	MP	MIOU
SDIF	40.426	33.053	42.269	20.510
PRI-Tran	37.358	29.009	37.722	16.757
BLSTM	50.658	46.201	46.406	30.912
BGRU	46.868	43.025	43.245	28.037
DCN	35.329	30.693	31.227	18.289
PR-RPAD	99.382	99.020	99.043	98.098

Bold values indicate the best performance.

recurrent neural network (BRNN) for DTOA sequence processing and connects the BRNN step output with a fully connected layer. The study employs the BGRU and BLSTM recurrent neural network (RNN) architectures, and it makes use of bidirectional recurrent neural network (TCN)s' popularity in sequence modeling [50] to achieve RSD. In this study, we choose the SDIF and PRI-Tran methods as the control experiments and establish the threshold value to finish the sequence search based on the literature. To determine the number of detected pulses throughout the search, the initial pulse is utilized as the beginning point; if it surpasses the threshold, it is the target pulse. Otherwise, the search fails. Then, the following pulse is used as the search's beginning point, and so on. As a result, the SDIF and PRI-Tran techniques' deinterleaving process does not need differentiating between known and unknown radars. The algorithm's closed-set and open-set indexes, however, ought to be determined using the known and unknown radar sets listed in the associated experimental parameter table.

4) *Performance Metrics:* ① *Closed-set scenario:* accuracy ($ACC_{closed-set}$), mean recall (MR), mean precision (MP), and mean intersection over union (MIOU).

② *Open-set scenario:* accuracy (ACC_{known}), and open-set classification rate (OSCR) [51].

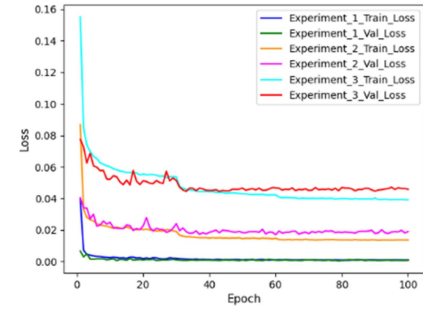
D. Results

- 1) The model performance analysis for Experiments 1–6 under the mixed dataset is displayed in Tables IX–XI. The loss versus accuracy curves in the model's training process from Experiments 1–3 are plotted as illustrated in Fig. 11 to examine the PR-RPAD

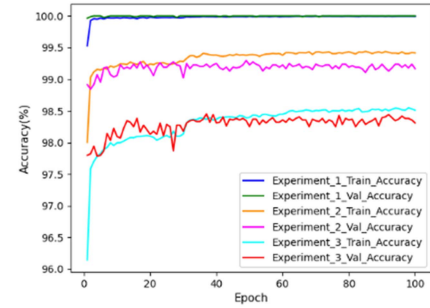
TABLE XI
Experiment 3: Analysis of Model Performance on a Mixed Dataset
(Closed Set)

Metrics Method	$ACC_{closed-set}$	MR	MP	MIOU
SDIF	38.413	23.613	30.010	14.365
PRI-Tran	37.383	21.721	41.519	12.537
BLSTM	43.934	31.752	35.504	20.323
BGRU	44.342	31.393	35.077	20.330
DCN	35.776	22.335	28.207	12.895
PR-RPAD	98.184	97.763	98.272	96.226

Bold values indicate the best performance.



(a)



(b)

Fig. 11. (a) and (b) Training loss versus accuracy curves for the PR-RPAD model under mixed datasets, closed-set scenario Experiments 1–3.

algorithm's overall performance and convergence in the closed-set situation. Among these, Fig. 11(a) displays the loss curve during the PR-RPAD model's training process; it is evident from the figure that the loss curve steadily declines as the number of training iterations rises and tends to stabilize at roughly 40 iterations. In Experiments 1–3, both stabilize at roughly 40 iterations, although the loss rises as the dataset's complexity increases. The accuracy curve during the PR-RPAD model's training process is depicted in Fig. 11(b); it is evident from the figure that the accuracy progressively rises as the number of training iterations grows, and the accuracy curve tends to stabilize at roughly 60 iterations. In Experiments 1–3, the accuracy rate has dropped as the dataset's complexity has increased, but all repetitions have stabilized at roughly 60 times. In conclusion, Experiments 1–3 of the closed-set scenario show

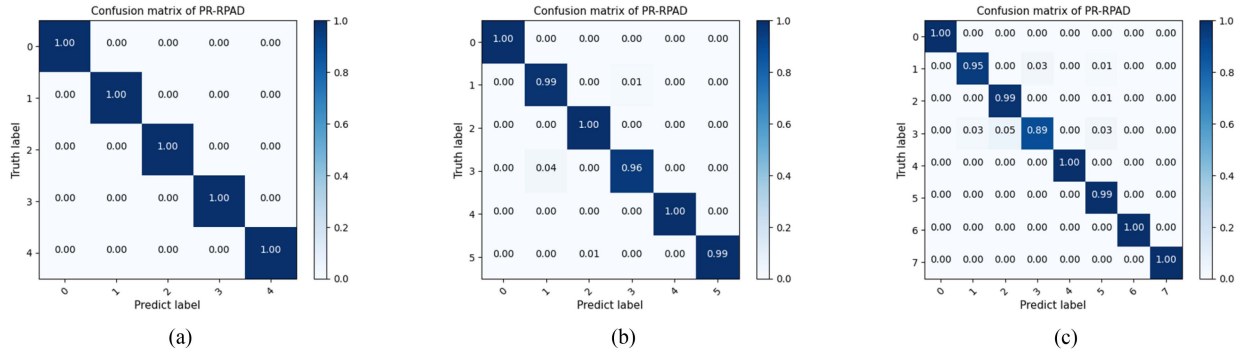


Fig. 12. (a)–(c) RSD confusion matrix for the mixed dataset of closed-set scenarios in Experiments 1–3.

that the accuracy rate declines and the loss of the PR-RPAD model grows as the dataset complexity increases; however, the curve tends to stabilize and the model converges at roughly 40–60 iterations on average.

As can be seen from Tables IX–XI, the model performance analysis results of Experiments 1–3 in the closed-set scenario, the PR-RPAD algorithm completely outperforms the traditional algorithms, such as SDIF, PRI-Tran, BLSTM, BGRU, and DCN, in the metrics of $ACC_{\text{closed-set}}$, MR, MP, and MIOU. One of the main causes of the comparison algorithm's poor performance is the lack of enough training data, and the quantity of training samples used in this work is not in line with previous research [19]. In addition, the PR-RPAD algorithm receives flawless scores on every index because there are not many radar radiation sources in Experiment 1. The PR-RPAD model performs well in terms of RSD and noise rejection, as seen in Fig. 12(a). As the dataset became more complicated in Experiment 2, the PR-RPAD model's deinterleaving performance declined. The PR-RPAD model continues to exhibit good performance in terms of partial radar radiation source deinterleaving and noise rejection, as illustrated in Fig. 12(b). For radar radiation sources 1 and 3 of the same model, with the same parameters of RF, PW, and PRI, the RSD features can only be extracted from the DOA and PA information. Due to the interleaving of the DOA and PA information of radar radiation sources 1 and 3, the RSD performance is slightly degraded. In Experiment 3, the PR-RPAD model's deinterleaving performance deteriorates much more as a result of the dataset's increased complexity. The PR-RPAD model continues to perform well in partial radar radiation source deinterleaving and noise rejection, as illustrated in Fig. 12(c). The radar type of radiation sources 1 and 3 remains the same, and the more radiation sources there are in the scenario, the more challenging RSD becomes. As a result, in Experiment 3, the deinterleaving performance for the identical model of radiation sources drastically declines.

Fig. 13 plots the loss versus accuracy curves for the model's training process for Experiments 4–6 in the open-set scenario. Among these, the loss curve during the PR-RPAD model's training process is depicted in Fig. 13(a);

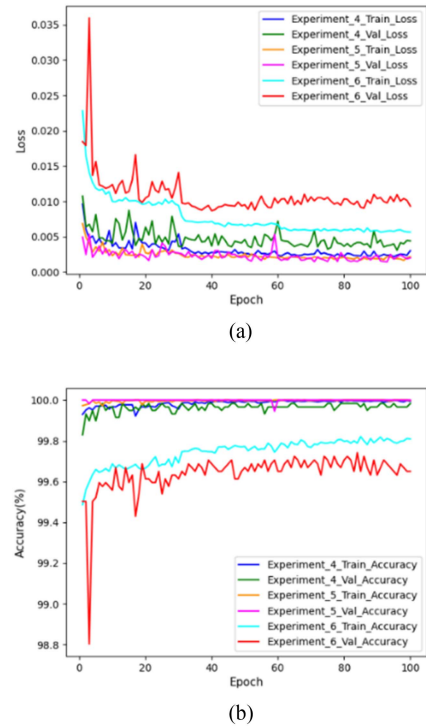


Fig. 13. (a) and (b) Training loss versus accuracy curves for the PR-RPAD model under mixed datasets, open-set scenario Experiments 4–6.

it is evident from the figure that the loss curve steadily declines as the number of training iterations increases and stabilizes at roughly 40 iterations. In Experiments 4–6, both stabilize at roughly 60 iterations, although the loss rises as the dataset's complexity increases. According to Fig. 13(b), which depicts the accuracy curve during the PR-RPAD model's training process, the accuracy progressively rises as the number of training iterations grows, and the accuracy curve tends to stabilize at roughly 40 iterations. The accuracy rate dropped in Experiments 4–6 as the dataset's complexity increased, but both iterations stabilized at roughly 60 times. The PR-RPAD model loss rose and the accuracy rate declined with increasing dataset complexity in Experiments 4–6 of the open-set scenario. However, after 40–60 iterations, the curve tends to settle and the model converges.

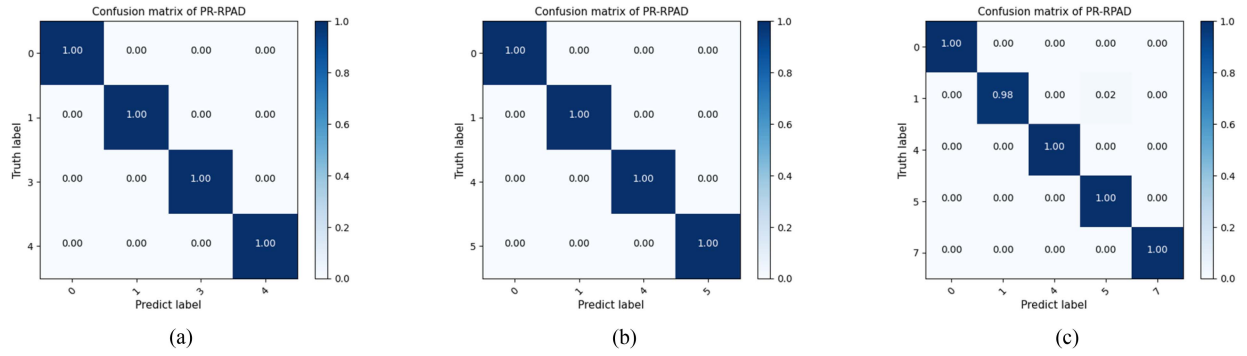


Fig. 14. (a)–(c) RSD confusion matrix for the mixed dataset of open-set scenarios in Experiments 4–6.

TABLE XII
Experiments 4–6: Model Performance Analysis With Mixed Datasets (Open Set)

Method		SDIF	PRI-Tran	PR-RPAD
Experiment 4	ACC_{known}	53.394	35.625	99.983
	OSCR	37.042	24.909	99.966
Experiment 5	ACC_{known}	50.316	37.441	100.000
	OSCR	38.487	29.121	82.391
Experiment 6	ACC_{known}	48.405	37.070	99.613
	OSCR	37.744	29.417	72.140

Bold values indicate the best performance.

In terms of ACC_{known} , OSCR, and other metrics, the PR-RPAD method substantially outperforms conventional algorithms like SDIF and PRI-Tran, according to the findings of model performance analysis for Experiments 4–6 in the open-set scenario, as shown in Table XII. Among these, the SDIF and PRI-Tran algorithms hunt for the regularity of the TOA sequence to complete RSD, which makes deinterleaving challenging in intricate PRI modulation settings. Meanwhile, the PR-RPAD algorithm has achieved flawless scores in all indexes because Experiment 4 had a limited number of radar radiation sources. The PR-RPAD model performs better in terms of noise rejection and RSD, as seen in Fig. 14(a). Experiment 5 shows that the PR-RPAD model performs better in known radar radiation source deinterleaving due to the dataset's increasing complexity, whereas its performance declines in unknown radar radiation source deinterleaving. As illustrated in Fig. 14(b), the PR-RPAD model continues to exhibit good RSD and noise rejection performance. As the complexity of the dataset increases in Experiment 6, the PR-RPAD model's deinterleaving performance deteriorates even more. The PR-RPAD model continues to exhibit good performance in terms of partial radar radiation source deinterleaving and noise rejection, as illustrated in Fig. 14(c). Radiation sources 1 and 5 interleave in terms of radiation parameters, which degrades the performance of the PR-RPAD model. In conclusion, the PR-RPAD model's performance steadily declines as the number of unknown radiation sources and parameter

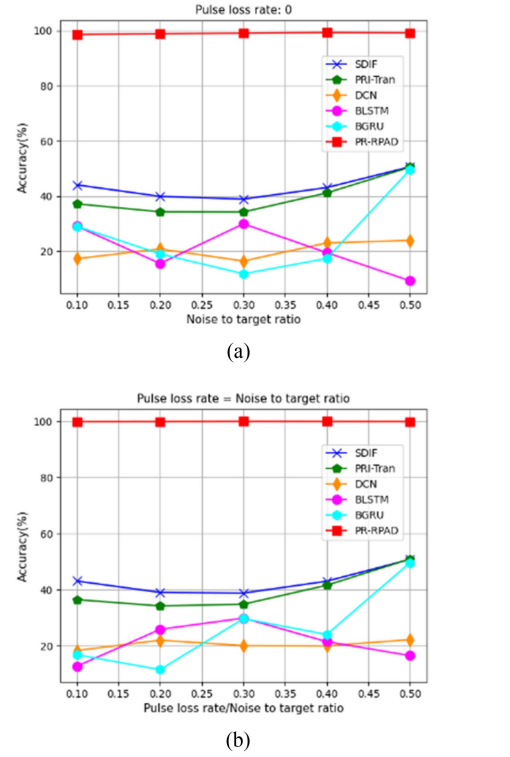


Fig. 15. RSD curves under specific interception conditions in Experiment 1. (a) Pulse loss rate equals zero. (b) Pulse loss rate equals the noise ratio.

complexity rises for the open-set scenario Experiments 4–6, and the OSCR index also declines.

- 2) Figs. 15–20 demonstrate the model performance analysis for Experiments 1–6 under particular interception scenario datasets. In the closed-set scenario, as illustrated in Figs. 15–17, the PR-RPAD algorithm outperforms both traditional techniques like SDIF and PRI-Tran and deep learning techniques like BLSTM, BGRU, and DCN in the case of a good reconnaissance measurement interception environment and a gradual deterioration of the reconnaissance measurement interception environment. It is also highly robust and unaffected by changes in the reconnaissance measurement interception environment.

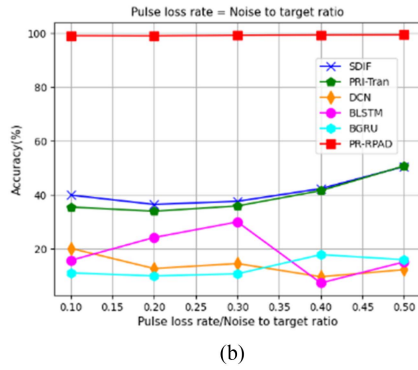
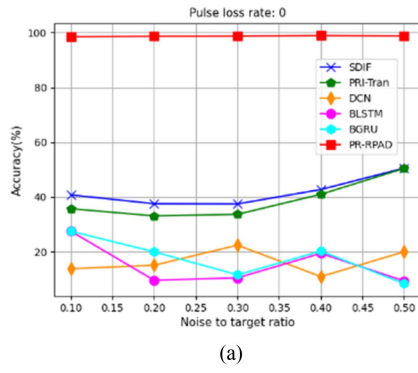


Fig. 16. RSD curves under specific interception conditions in Experiment 2. (a) Pulse loss rate equals zero. (b) Pulse loss rate equals the noise ratio.

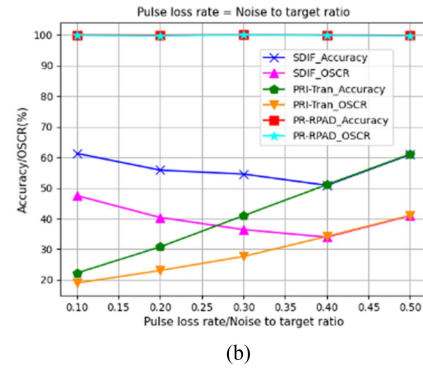
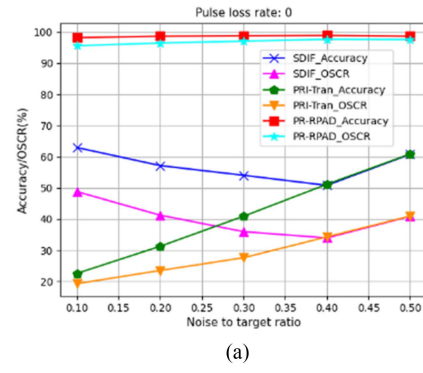


Fig. 18. RSD curves under specific interception conditions in Experiment 4. (a) Pulse loss rate equals zero. (b) Pulse loss rate equals the noise ratio.

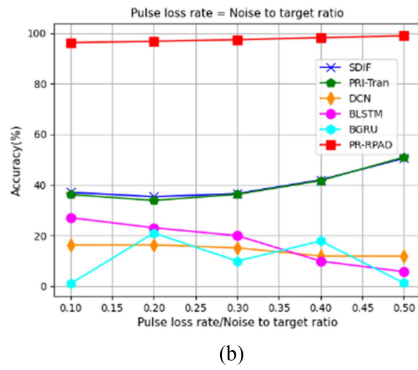
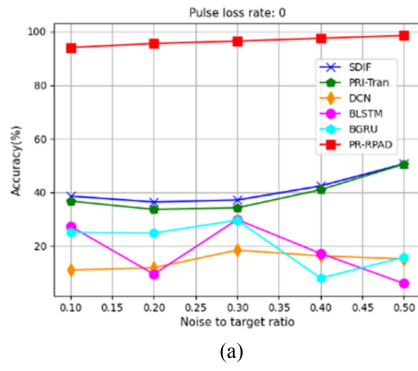


Fig. 17. RSD curves under specific interception conditions in Experiment 3. (a) Pulse loss rate equals zero. (b) Pulse loss rate equals the noise ratio.

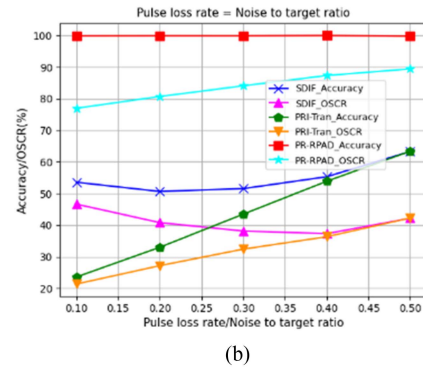
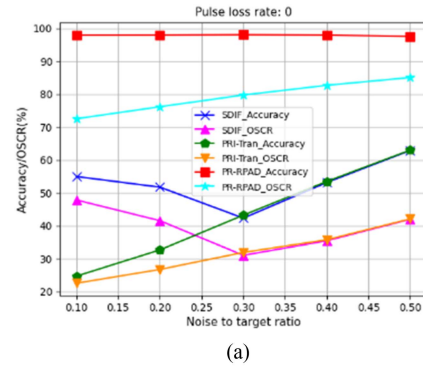


Fig. 19. RSD curves under specific interception conditions in Experiment 5. (a) Pulse loss rate equals zero. (b) Pulse loss rate equals the noise ratio.

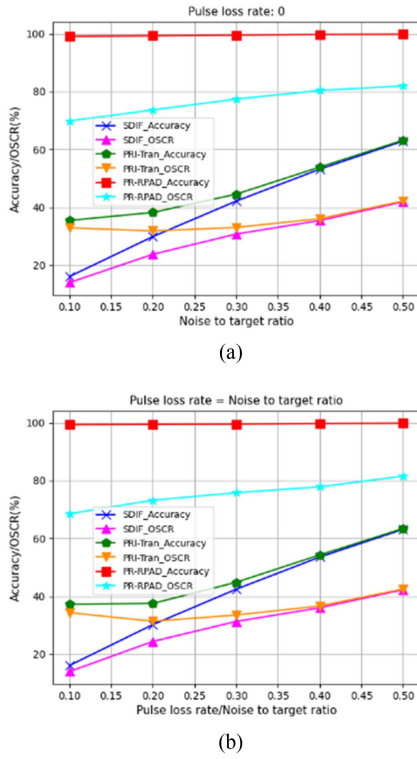


Fig. 20. RSD curves under specific interception conditions in Experiment 6. (a) Pulse loss rate equals zero. (b) Pulse loss rate equals the noise ratio.

Meanwhile, we believe that the study of the deinterleaving algorithm's performance should compute its capacity to differentiate between noise, clutter, and interference signals in addition to assessing its deinterleaving performance to the target radiation source pulse. As a result, in conventional RSD algorithms like SDIF and PRI-Tran, the noise in the radiation pulse stream gradually increases as the reconnaissance measurement interception environment changes. This results in a diluted density of the target radiation source pulse, which is then correctly extracted with the help of the algorithm's strong noise rejection capability, and an overall upward trend in RSD performance. The performance curve displays an up-and-down phenomenon, and the noise in the radiation pulse stream gradually increases with the change of the reconnaissance measurement interception environment in deep learning RSD algorithms like BLSTM, BGRU, DCN, etc. This leads to unstable RSD performance, poor robustness, and an increased risk of deinterleaving.

In the open-set scenario, as illustrated in Figs. 18–20, the PR-RPAD algorithm outperforms traditional techniques like SDIF and PRI-Tran in known radar radiation source deinterleaving under both good and gradually deteriorating reconnaissance measurement interception environments. It also exhibits strong robustness and is not affected by changes in the reconnaissance measurement interception environment. The PR-RPAD model's performance in Experiment 4 is stable with minimal fluctuation in the data from an unknown radar radiation source. In Experiments 5 and 6, the noise in the radiation pulse stream gradually increases as the reconnaissance measurement interception

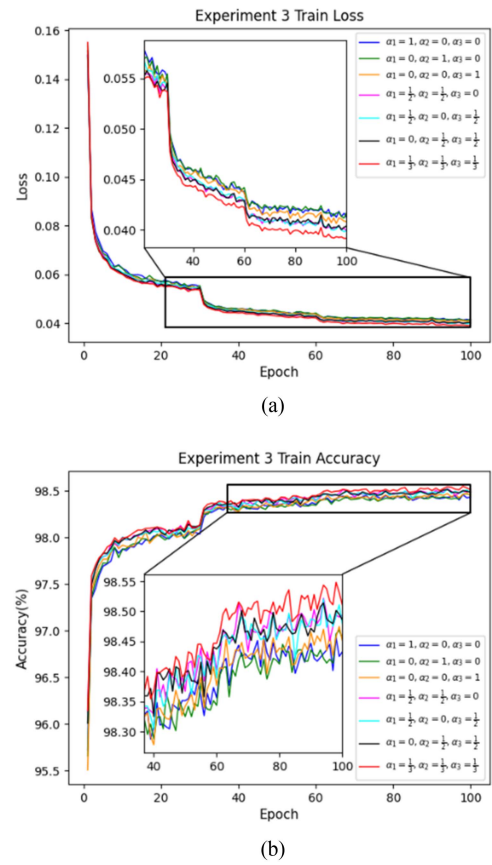


Fig. 21. (a) and (b) Experiment 3: PR-RPAD model training loss and accuracy curves with varying weight values.

environment changes, but the total amount of data stays the same. Furthermore, the description parameters for the target radiation source pulse and noise differ significantly. The PR-RPAD algorithm's strong noise discrimination capability supports the target radiation source pulses' proper deinterleaving and the overall RSD accuracy's upward trend. In SDIF and PRI-Tran, as the reconnaissance measurement intercept environment changes, i.e., the proportion of noise increases, leading to DTOA anomalies; at the same time, the loss rate increases, leading to a decrease in radiation pulse density. The algorithms SDIF and PRI-Tran mine the potential laws of TOA sequences to gradually remove noise and dilute the intercepted sequences. Finally, the target radar sequence is extracted from the low-density intercept sequence to achieve the overall improvement of RSD accuracy.

- 3) The impact of the loss function weights on the PR-RPAD model's performance is examined using the loss function weights α_1 , α_2 , and α_3 of $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, $(1/2, 1/2, 0)$, $(1/2, 0, 1/2)$, $(0, 1/2, 1/2)$, and $(1/3, 1/3, 1/3)$, respectively, to examine the RSD performance of the PR-RPAD model in Experiments 3 and 6. Fig. 21 displays the PR-RPAD model's training loss versus accuracy curves for Experiment 3 using the weights mentioned earlier. The loss curve for weight $\alpha_1 = \alpha_2 = \alpha_3 = 1/3$ converges more quickly than

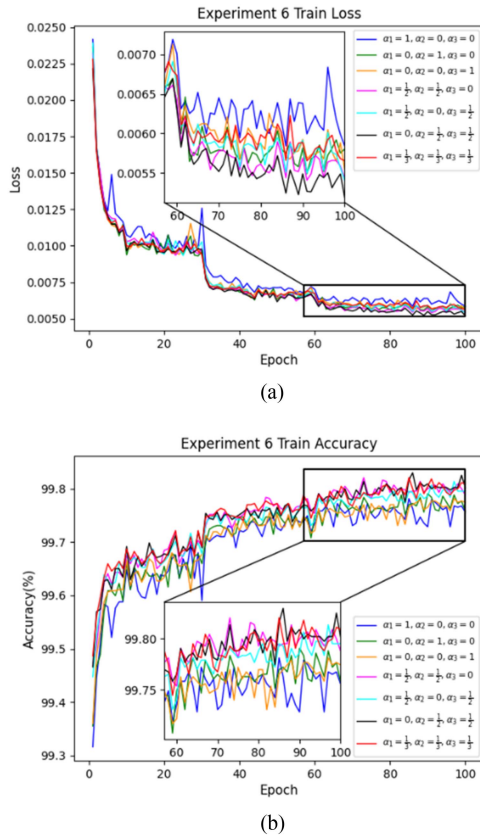


Fig. 22. (a) and (b) Experiment 6: PR-RPAD model training loss and accuracy curves with varying weight values.

the other cases in Fig. 21(a). The accuracy curve for weight $\alpha_1 = \alpha_2 = \alpha_3 = 1/3$ is higher than that of other cases in Fig. 21(b). Thus, the PR-RPAD model's RSD performance may be maximized by adjusting the loss weight $\alpha_1 = \alpha_2 = \alpha_3 = 1/3$ on various pyramid-level feature reconstruction outcomes.

Fig. 22 displays the PR-RPAD model's training loss versus accuracy curves for Experiment 6 using the weights mentioned above. The pace of convergence of the PR-RPAD model loss curves is not significantly impacted by the various weight choices in the open-set scenario, as shown in Fig. 22(a). Compared to the other cases, the PR-RPAD model with weights α_1, α_2 , and α_3 that take the values of $(1/2, 1/2, 0)$, $(1/2, 0, 1/2)$, $(0, 1/2, 1/2)$, and $(1/3, 1/3, 1/3)$ has a greater unintertwining accuracy curve in Fig. 22(b). Thus, the deinterleaving performance of the PR-RPAD model can be enhanced by reconstructing and fusing the characteristics of the various pyramidal layers.

- 4) In open-set scenarios, the effect of CS generation training on the PR-RPAD model's RSD performance is examined. Fig. 23 displays the PR-RPAD model training loss versus accuracy curves for Experiments 4–6. In PR-RPAD, the CS generation training module is not introduced during model training, while in PR-RPAD+CS, the CS generation training module is introduced during model training. The figure shows that during the training of Experiments 4 and

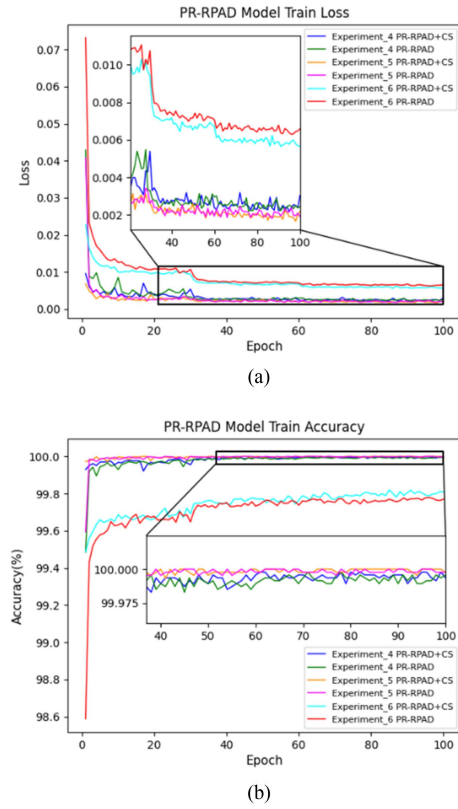


Fig. 23. Experiment 4–6: Training loss versus accuracy curve for the PR-RPAD model.

5, the loss and accuracy curves of PR-RPAD+CS and PR-RPAD overlap, making it challenging to determine how much the CS generation training module contributed to the RSD performance. The PR-RPAD+CS loss curves in Experiment 6 converge more quickly than PR-RPAD and are stable around 40–60 iterations when the number of radar radiation sources increases and the radiation parameters become more complex. In the meantime, the PR-RPAD+CS RSD performance curve outperforms PR-RPAD. To sum up, it can be said that as the number of radar radiation sources and radiation parameter complexity increase, the CS generation training module's contribution to improving RSD performance becomes more noticeable. This is particularly evident in Experiment 6.

- 5) To further demonstrate the superiority of the PR-RPAD method proposed in this article, an ablation study is conducted to verify the function and benefits of the algorithm's Laplacian pyramid feature reconstruction and fusion module in the deinterleaving of radar signals. Using the data from Experiments 3 and 6, ablation experiments are designed. The Laplacian PR module's inclusion or absence in the method is controlled to investigate the impact on the RSD performance. Among these, reciprocal point adversarial deinterleaving (RPAD) is the algorithm that does not need the PR module. In the ablation experiment, the model's deinterleaving performance on the mixed

TABLE XIII
Analysis of the Model's Performance in the Ablation Studies via Mixed Datasets for Experiments 3 (Closed Set) and 6 (Open Set)

Method		RPAD	PR-RPAD
Metrics			
Experiment 3	$ACC_{closed-set}$	91.526	98.184
	MR	87.228	97.763
	MP	91.697	98.272
	MIOU	81.540	96.226
Experiment 6	ACC_{known}	94.935	99.613
	OSCR	32.227	72.140

Bold values indicate the best performance.

TABLE XIV
Analysis of the Model's Performance in the DTOA Control Experiments via Mixed Datasets for Experiments 3 (Closed Set) and 6 (Open Set)

Input Data		PDG-DTOA	PDG
Metrics			
Experiment 3	$ACC_{closed-set}$	91.197	98.184
	MR	84.795	97.763
	MP	91.014	98.272
	MIOU	79.967	96.226
Experiment 6	ACC_{known}	92.135	99.613
	OSCR	62.526	72.140

Bold values indicate the best performance.

datasets of Experiments 3 and 6 is displayed in Table XIII. The table illustrates that the application of the Laplacian pyramid feature reconstruction and fusion architecture can achieve tower reshaping of the PDG features from low to high resolution, extend the different resolution characterization of the PDG edges and texture features, and improve the deinterleaving performance of radar-radiated source signals in both the closed-set scenario (Experiment 3) and the open-set scenario (Experiment 6).

- 6) The PR-RPAD algorithm's performance analysis for deinterleaving when DTOA data is unavailable or missing. Taking into account RSD's real-time requirements, the internal storage capacity limitations following PDG synthesis, and the potential for adverse effects from nearby pulses, a PDG is generated from a single PDW when $len_{window} = 1$ is chosen. Only the TOA difference between neighboring pulses is used in the transformation procedure mentioned earlier; the temporal correlations between other pulses are not investigated. Consequently, it is possible to adjust whether or not the PDG has DTOA information to confirm its impact on the deinterleaving performance of radar signals; the PDG that does not have DTOA information is known as PDG-DTOA. When PDG-DTOA and PDG are input, respectively, the model deinterleaving performance results under the mixed dataset for Experiments 3 and 6 are displayed in Table XIV. The table shows that, in both the closed-set scenario (Experiment 3) and the open-set scenario (Experiment 6), DTOA

somewhat enhances the radar-radiated source signal deinterleaving performance. In addition, when DTOA data are absent or unavailable, the PR-RPAD method may nevertheless guarantee the regular operation of the deinterleaved process, unlike the conventional approach that solely uses TOA information for deinterleaving.

VI. CONCLUSION

In this study, we propose an open-set deinterleaving technique for radar radiation source signals, called PR-RPAD, which combines reciprocal point adversarial learning with Laplacian PR. The PR-RPAD algorithm employs the visual characterization and feature enhancement of the PDW to the PDG through the symmetric mapping of grayscale matrices and DCCI_Otsu image enlargement algorithms. Employing the Laplacian pyramid model of multiresolution feature reconstruction and fusion to tower reshape PDG features from low to high resolution. In parallel, the concept of reciprocal points is introduced to model the open space, and an instantiated enhancement method is proposed. The deinterleaving of the known and unknown radar radiation sources in the open space is finished under the counter mechanism of reciprocal points and the known radar. The following conclusions can be made in light of the studies mentioned earlier.

- 1) This article has examined the deinterleaving of signals from known and unknown radar radiation sources using single-function, multifunction, and the same type, as well as radar sources using common PRI modulation styles (such as jitter PRI).
- 2) Unlike the conventional RSD approaches [5], [14], [15], the PR-RPAD algorithm can handle complex PRI modulation styles and searches for sequences without determining the PRI or PRF. Many search and merging rounds are not necessary for radar radiation sources that have many pulses in a cycle.
- 3) The PR-RPAD algorithm can deinterleave signals from multiple radar radiation sources using a single network, as opposed to the RSD methods of neural networks and automata [4], [16], [17], [18]. This eliminates the need to train a different network for each radar radiation source target and iteratively iterates inputs and outputs to the radar source data.
- 4) In comparison to the SDIF, PRI-Tran, and SSD (BLSTM, BGRU, and DCN) [19] algorithms, the PR-RPAD algorithm is simple to train and converge, and it exhibits notable performance advantages, robustness, and strong environmental adaptability for known and unknown radar radiation source deinterleaving.
- 5) When pulse parameters are not complete, the PR-RPAD method nevertheless performs reasonably well in terms of deinterleaving. Nevertheless, PDG creation, amplification, and enhancement processing

result in higher algorithmic complexity and deinterleaving time cost, indicating the direction for future algorithm optimization.

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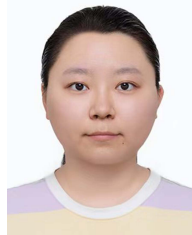
Yang-Yang Dong (Member, IEEE) received the B.Eng. and Ph.D. degrees in electronic science and technology from Xidian University, Xi'an, China, in 2012 and 2017, respectively.

Since 2017, he has been a Lecturer and then an Associate Professor with the School of Electronic Engineering, Xidian University. His research interests include array signal processing and intelligent spectrum sensing.



Chunxi Dong was born in Sanmenxia, Henan, China. He received the Ph.D. degree in electronic science and technology from Xidian University, Xi'an, China, in 2004.

He is currently a Professor and the Dean of the Department of Information Technology, School of Electronic Engineering, Xidian University. His research interests include high-speed signal processing and system simulation.



Ronghua Guo received the B.Eng. degree in information countermeasure technology in 2020 from Xidian University, Xi'an, China, where she is currently working toward the Ph.D. degree in electronic science and technology.

Her current research interests include pulse-splitting detection and radar emitter deinterleaving.



Wenbo Li received the B.Eng. degree in information countermeasure technology from Shenyang Ligong University, Shenyang, China, in 2020. He is currently working toward the Ph.D. degree in electronic science and technology with Xidian University, Xi'an, China.

His current research interests include deep learning, radar emitter deinterleaving, and classification.



Zhiyuan Li received the B.Eng. degree in communication engineering from Dalian University, Dalian, China, in 2020. He is currently working toward the Ph.D. degree in electronic science and technology with Xidian University, Xi'an, China.

His current research interests include deep learning and radar mode identification.